

ME 616: Convective Heat and Mass Transfer

Course Code	ME 616
Course Name	Convective Heat and Mass Transfer
L-T-P-C	3-0-0-3
Prerequisites	ME-210: Fluid Mechanics, ME-303: Heat Transfer, IC110: Engineering Mathematics
Intended for	UG/PG
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Governing Equations**
Continuity, Momentum and Energy Equations, reduction of equations for various fluid flow systems, boundary layer approximations to momentum and energy, scale analysis, Introduction to nano-heat transfer. [6 Lectures]
- **Laminar external flow and heat transfer**
Scale analysis, similarity solutions for flat plate (Blasius solution), scale analysis of thick and thin thermal boundary layer, Integral method solutions for flow over an isothermal flat plate, flat plate with constant heat flux and with varying surface temperature, flows with pressure gradient. [6 Lectures]
- **Laminar internal flow and heat transfer**
(a) Exact solutions to N-S equations for flow through channels and circular pipe, Fully developed forced convection in pipes with different wall boundary conditions, Forced convection in the thermal entrance region of ducts and channels (Graetz solution), heat transfer in the combined entrance region, (b) Integral method for internal flows with different wall boundary conditions. [8 Lectures]
- **Natural convection heat transfer**
Governing equations for natural convection, Boussinesq approximation, Scale analysis: thermal and hydrodynamic boundary layers, Scale analysis in flow in vertical plate, Walls different boundary conditions: constant temperature and heat flux, Similarity and integral solutions, effects of inclination, Natural convection in enclosures, mixed convection heat transfer past vertical plate and in enclosures. [10 Lectures]
- **Turbulent convection**
Governing equations for averaged turbulent flow field (RANS), Analogies between heat and Mass transfer (Reynolds, Prandtl-Taylor and von Karman Analogies), Turbulence Models (Zero, one and two equation models), Turbulent flow and heat transfer across flat plate and circular tube, Turbulent natural convection heat transfer, Empirical correlations for different configurations. [6 Lectures]
- **Convective mass transfer**
Mass conservation, mass diffusivities, laminar forced convection, internal forced convection, natural convection: mass and heat transfer driven flows, turbulent flows: time averaged concentration equation, effect of chemical reaction, concept of boiling and condensation phenomena. [6 Lectures]

Text Books

1. Adrian Bejan, **Convective Heat Transfer**, 4th Edition, John Wiley & Sons, 2013.
2. Kays W M and Crawford M E, **Convective Heat and Mass Transfer**, 3rd Edition, McGraw Hill Int Edition, 1993.
3. Spalding D B, **Introduction to Convective Mass Transfer**, McGraw Hill, 1963.

References

1. Louis C. Burmeister, **Convective Heat Transfer**, 2nd Edition, Wiley-Interscience, 1993
2. Arpaci & Larcen, **Convective Heat Transfer**, Prentice Hall, 1984.
3. Sadik Kakac, Yaman Yener, Anchasa Pramuanjaroenkij, **Convective Heat Transfer**, 3rd Edition, CRC Press, 2013.
4. Bird R. B., Stewart W. E. and Lightfoot E. N., **Transport Phenomena**, 1st Edition, John Wiley and sons, Inc., 1960.
5. Schlichting H., **Boundary Layer Theory**, 6th Edition, McGraw Hill, 1968.